

Newton-Cotes Formulas of Closed Type

Newton-Cotes formula is a **closed-form** expression constructed by integrating of **Lagrange interpolating polynomials** to approximate a **definite integral**.

$$\begin{aligned} \int_a^b f(x) \cdot dx &\approx \int_a^b p_n(x) \cdot dx = \int_a^b \left[\sum_{k=0}^n L_k^{(n)}(x) \cdot f(x_{i+k}) \right] \cdot dx = \sum_{k=0}^n \left[\left(\int_a^b L_k^{(n)}(x) \cdot dx \right) \cdot f(x_{i+k}) \right] \\ &= \sum_{k=0}^n w_k \cdot f(x_{i+k}) \end{aligned}$$

Newton-Cotes Formulas of Closed Type: $n \geq 1, a = x_i, b = x_{i+n}$

$$\int_{x_i}^{x_{i+n}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+n}} p_n(x) \cdot dx = \int_{x_i}^{x_{i+n}} \left[\sum_{k=0}^n L_k^{(n)}(x) \cdot f(x_{i+k}) \right] \cdot dx = \sum_{k=0}^n w_k \cdot f(x_{i+k})$$

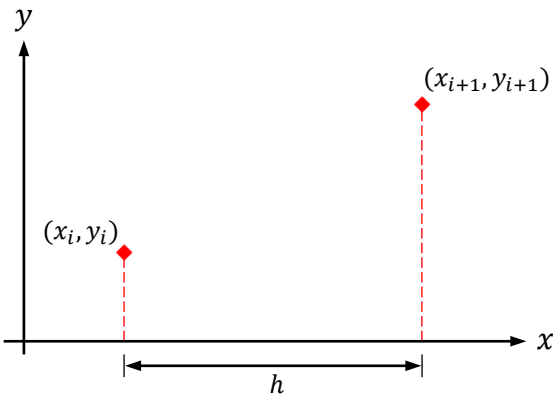
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The Trapezoidal Rule

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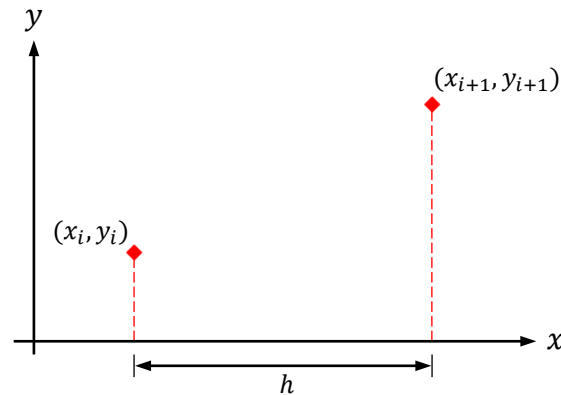


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$$n = 1$$

$$f(x) \approx p_1(x)$$



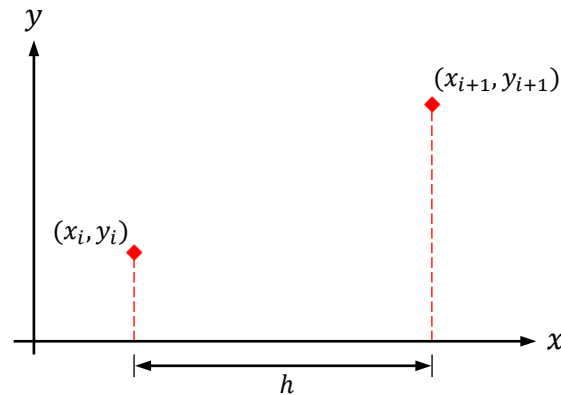
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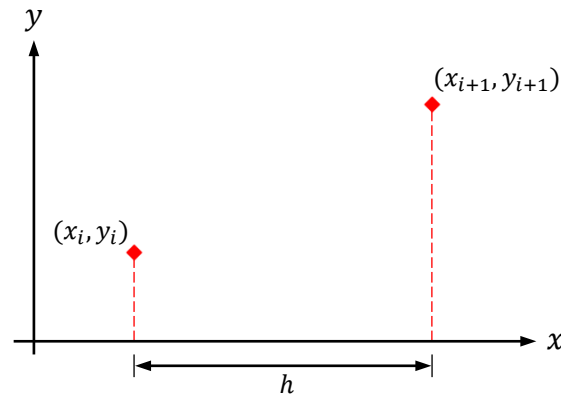
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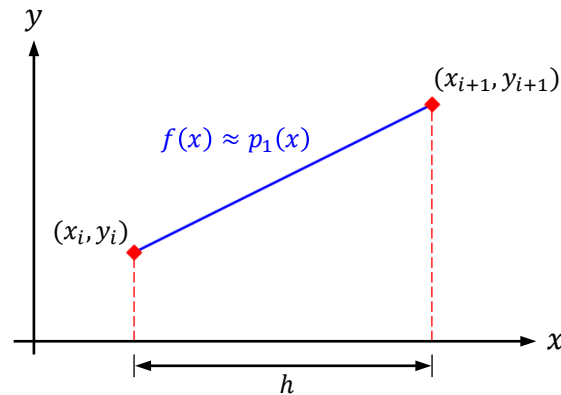
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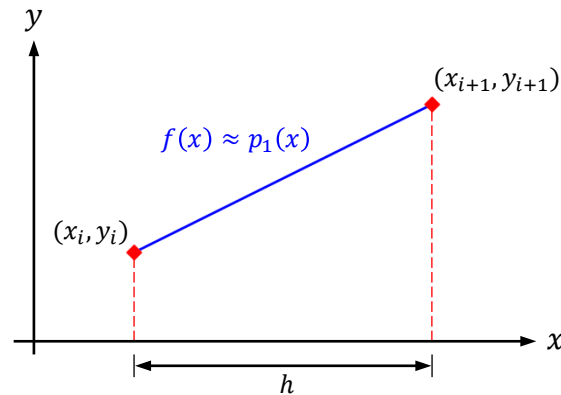
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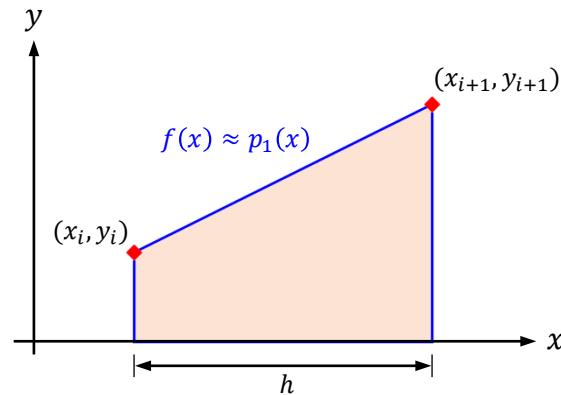
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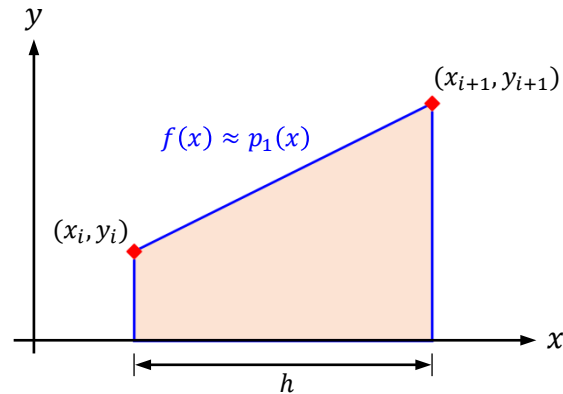
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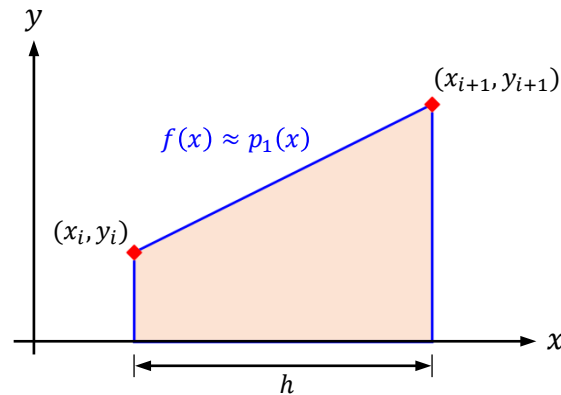
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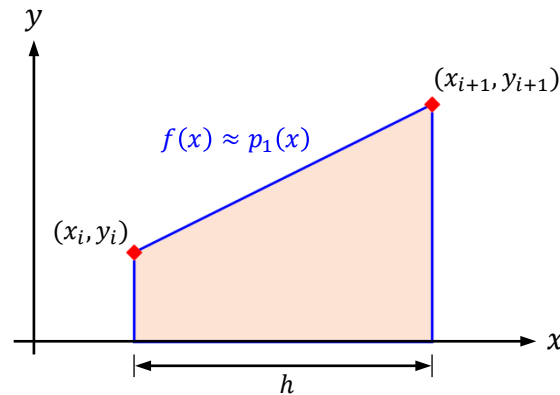
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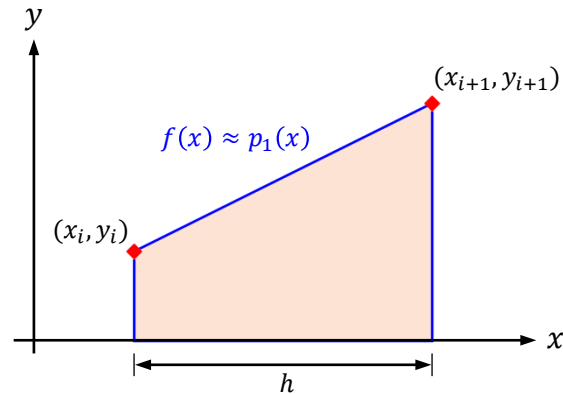
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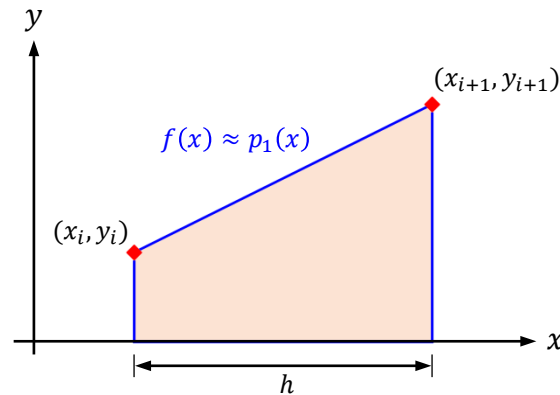
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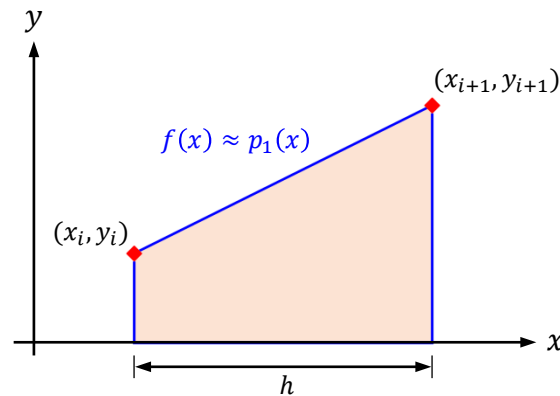
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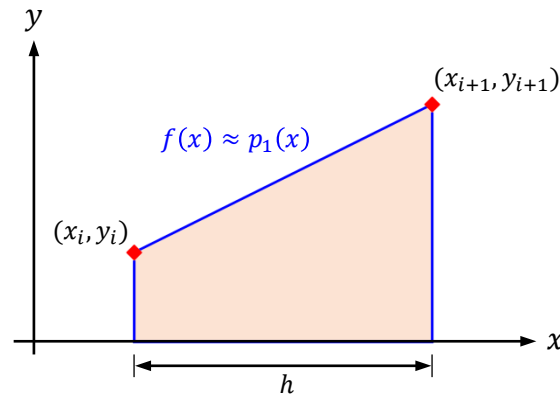
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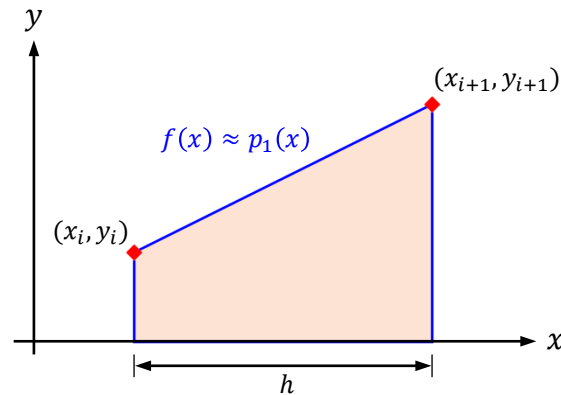
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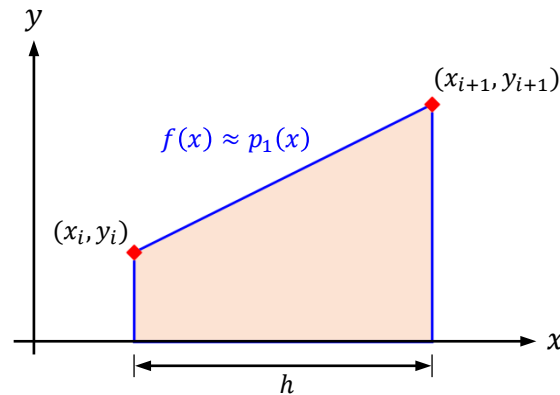
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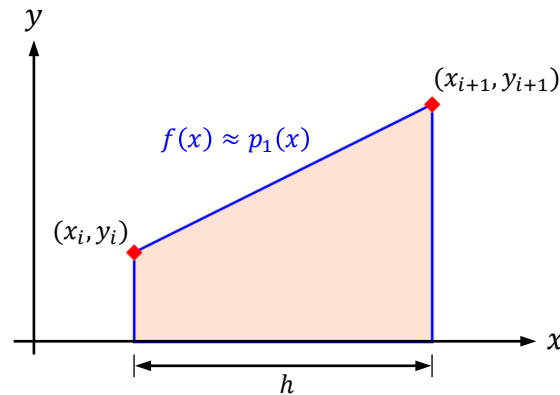
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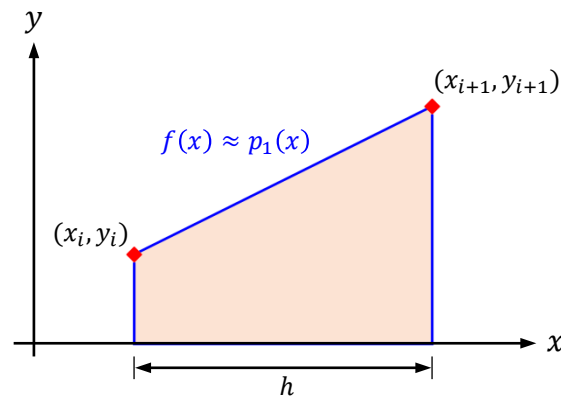
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The Trapezoidal Rule: The Local Truncation Error

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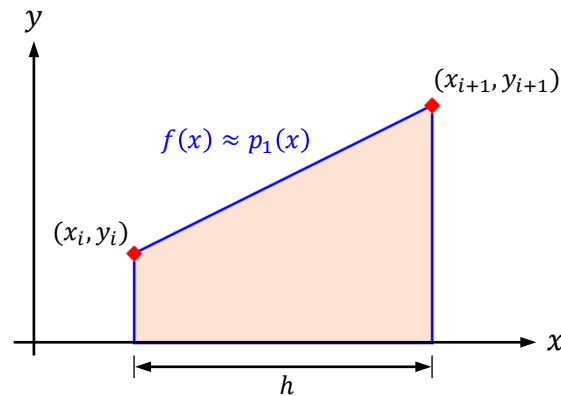
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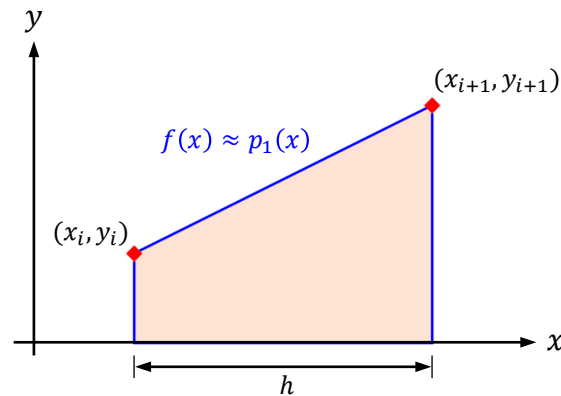


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$$\begin{aligned} \int_{x_i}^{x_{i+1}} f(x) \cdot dx &= (x_{i+1} - x_i) \cdot f(x_i) + \frac{(x_{i+1} - x_i)^2}{2!} \cdot f'(x_i) \\ &+ \frac{(x_{i+1} - x_i)^3}{3!} \cdot f''(x_i) + \frac{(x_{i+1} - x_i)^4}{4!} \cdot f^{(3)}(x_i) + \dots \end{aligned}$$

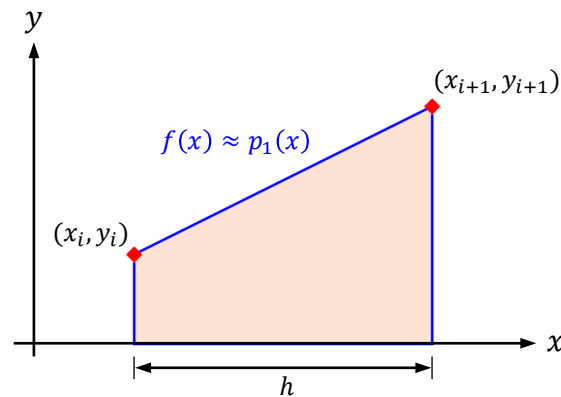


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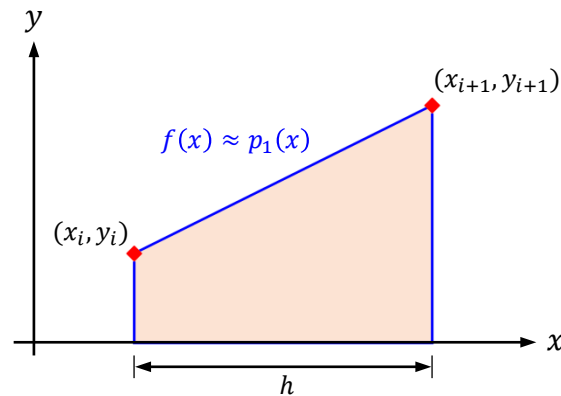
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$$R_{[x_i, x_{i+1}]}^{(1)} = \int_{x_i}^{x_{i+1}} f(x) \cdot dx - \int_{x_i}^{x_{i+1}} p_1(x) \cdot dx$$



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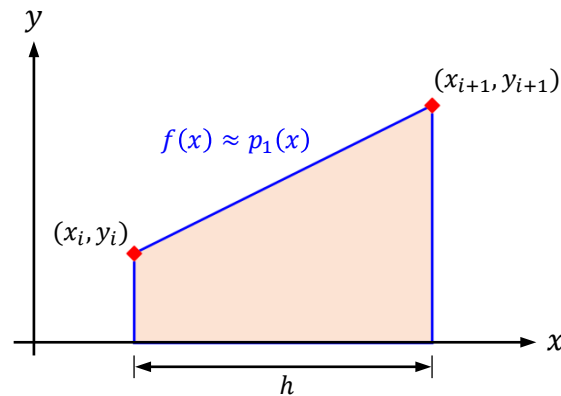
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$$R_{[x_i, x_{i+1}]}^{(1)} = \int_{x_i}^{x_{i+1}} f(x) \cdot dx - \int_{x_i}^{x_{i+1}} p_1(x) \cdot dx = h \cdot f(x_i) + \frac{h^2}{2} \cdot f'(x_i) + \frac{h^3}{6} \cdot f''(x_i) + \frac{h^4}{24} \cdot f^{(3)}(x_i) + \dots$$

$$- \frac{h}{2} \cdot f(x_i) - \frac{h}{2} \cdot f(x_{i+1})$$



Newton-Cotes Formulas of Closed Type

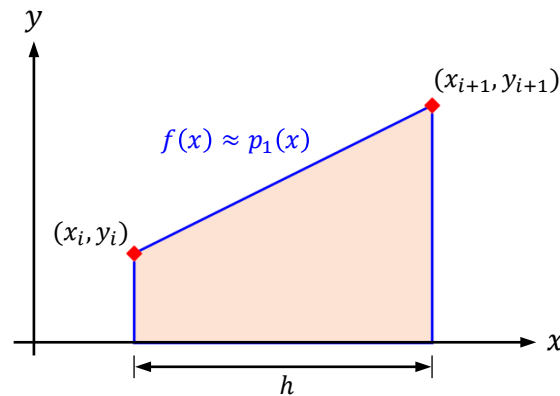
The Trapezoidal Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

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$$- \frac{h}{2} \cdot f(x_i) - \frac{h}{2} \cdot \left[f(x_i) + h \cdot f'(x_i) + \frac{h^2}{2} \cdot f''(x_i) + \frac{h^3}{6} \cdot f^{(3)}(x_i) + \dots \right]$$



Newton-Cotes Formulas of Closed Type

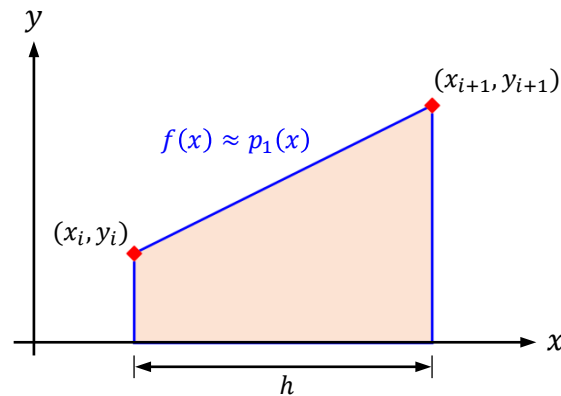
The Trapezoidal Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

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$$- h \cdot f(x_i) - \frac{h^2}{2} \cdot f'(x_i) - \frac{h^3}{4} \cdot f''(x_i) - \frac{h^4}{12} \cdot f^{(3)}(x_i) - \dots$$



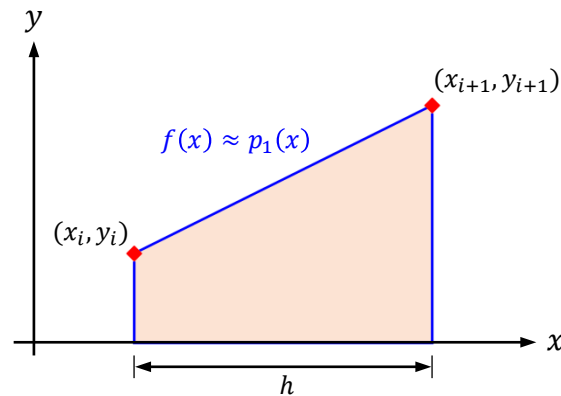
Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

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$$R_{[x_i, x_{i+1}]}^{(1)} = \int_{x_i}^{x_{i+1}} f(x) \cdot dx - \int_{x_i}^{x_{i+1}} p_1(x) \cdot dx = -\frac{h^3}{12} \cdot f''(x_i) - \frac{h^4}{24} \cdot f^{(3)}(x_i) + \dots$$



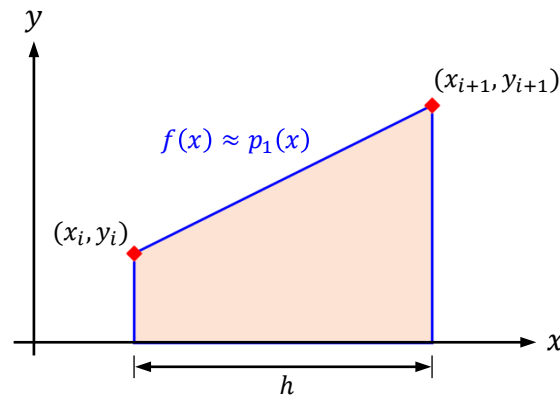
Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Local Truncation Error

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$$\int_{x_i}^{x_{i+1}} f(x) \cdot dx = h \cdot f(x_i) + \frac{h^2}{2} \cdot f'(x_i) + \frac{h^3}{6} \cdot f''(x_i) + \frac{h^4}{24} \cdot f^{(3)}(x_i) + \dots$$

$$\begin{aligned} R_{[x_i, x_{i+1}]}^{(1)} &= \int_{x_i}^{x_{i+1}} f(x) \cdot dx - \int_{x_i}^{x_{i+1}} p_1(x) \cdot dx = -\frac{h^3}{12} \cdot f''(x_i) - \frac{h^4}{24} \cdot f^{(3)}(x_i) + \dots \\ &= -\frac{h^3}{12} \cdot f''(\xi_{i+1}); \quad x_i \leq \xi_{i+1} \leq x_{i+1} \end{aligned}$$



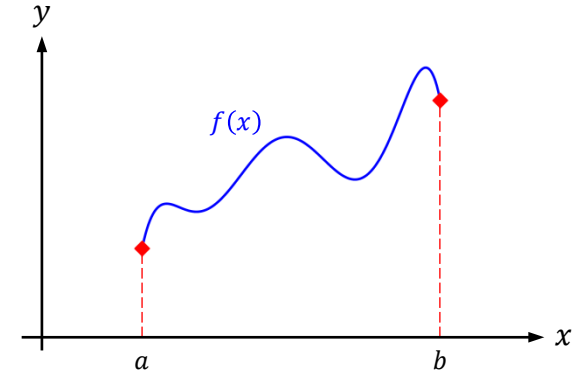
Newton-Cotes Formulas of Closed Type

Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

Newton-Cotes Formulas of Closed Type

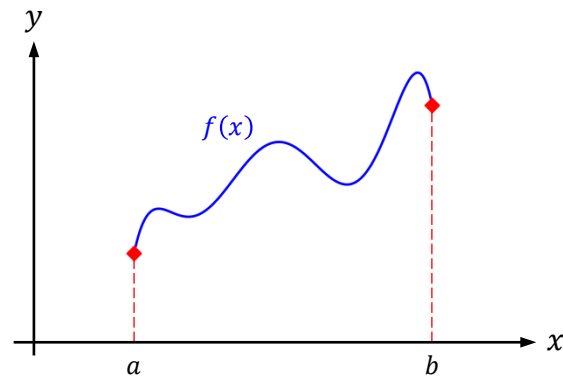
The Trapezoidal Rule: The Global Truncation Error



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

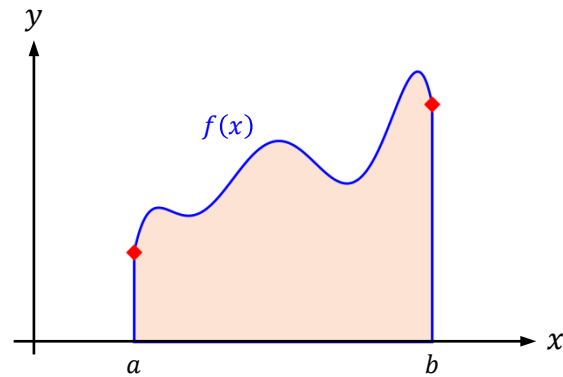
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

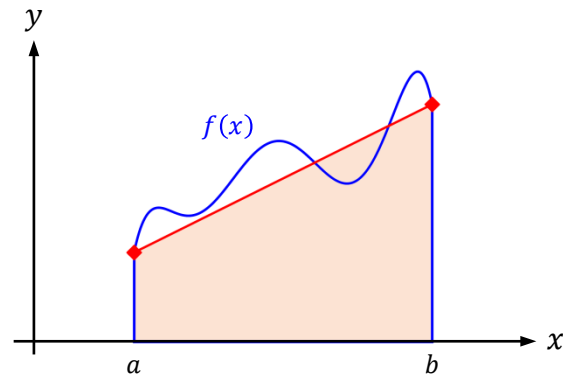
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

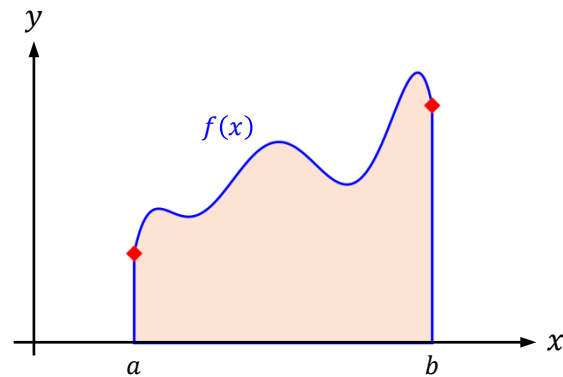
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

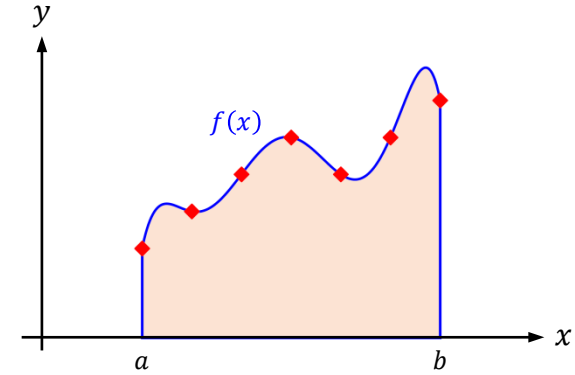
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

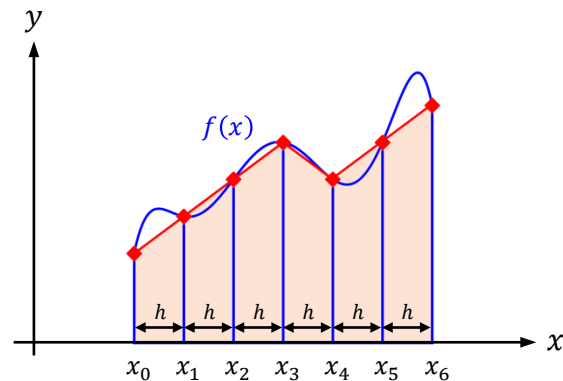
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

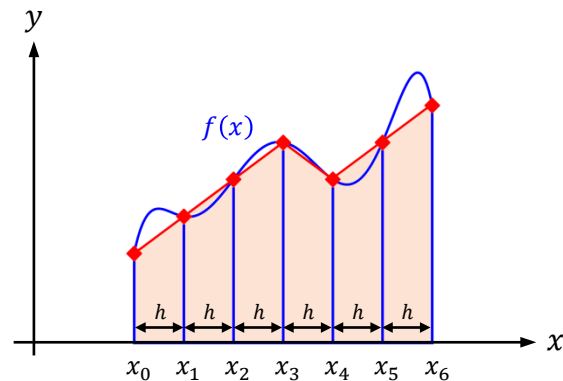
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+h} f(x) \cdot dx + \int_{a+h}^{a+2h} f(x) \cdot dx + \cdots + \int_{b-2h}^{b-h} f(x) \cdot dx + \int_{b-h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_1} f(x) \cdot dx + \int_{x_1}^{x_2} f(x) \cdot dx + \cdots + \int_{x_{N-2}}^{x_{N-1}} f(x) \cdot dx + \int_{x_{N-1}}^{x_N} f(x) \cdot dx \end{aligned}$$

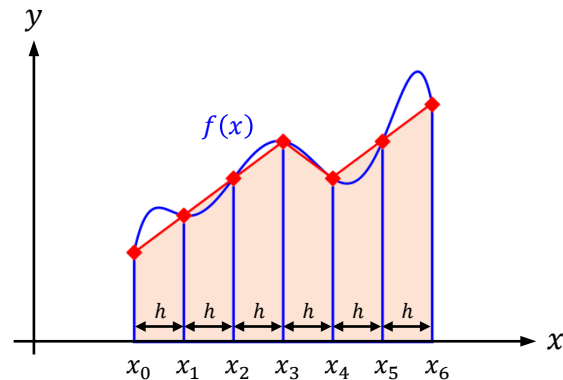


Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+h} f(x) \cdot dx + \int_{a+h}^{a+2h} f(x) \cdot dx + \cdots + \int_{b-2h}^{b-h} f(x) \cdot dx + \int_{b-h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_1} f(x) \cdot dx + \int_{x_1}^{x_2} f(x) \cdot dx + \cdots + \int_{x_{N-2}}^{x_{N-1}} f(x) \cdot dx + \int_{x_{N-1}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b - a}{N}$$

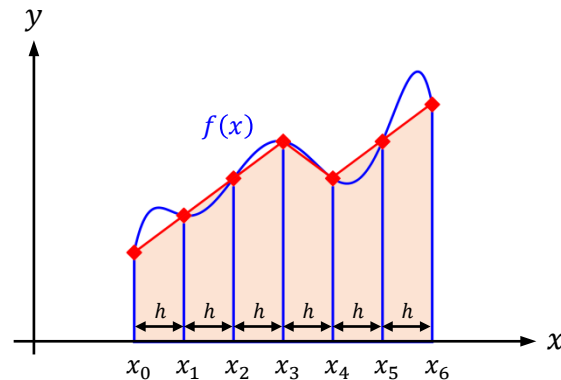


Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+h} f(x) \cdot dx + \int_{a+h}^{a+2h} f(x) \cdot dx + \cdots + \int_{b-2h}^{b-h} f(x) \cdot dx + \int_{b-h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_1} f(x) \cdot dx + \int_{x_1}^{x_2} f(x) \cdot dx + \cdots + \int_{x_{N-2}}^{x_{N-1}} f(x) \cdot dx + \int_{x_{N-1}}^{x_N} f(x) \cdot dx \end{aligned}$$

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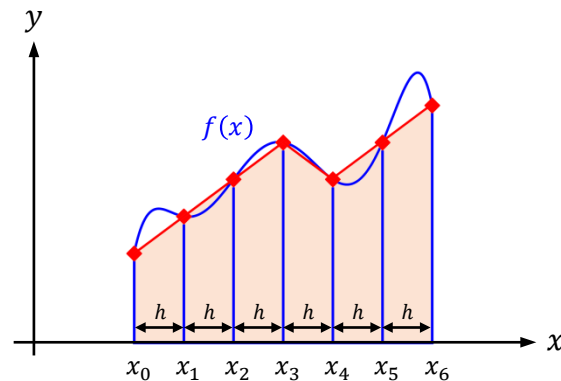
Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

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$$R_{[a,b]}^{(1)} = R_{[x_0,x_1]}^{(1)} + R_{[x_1,x_2]}^{(1)} + \cdots + R_{[x_{N-1},x_N]}^{(1)}$$



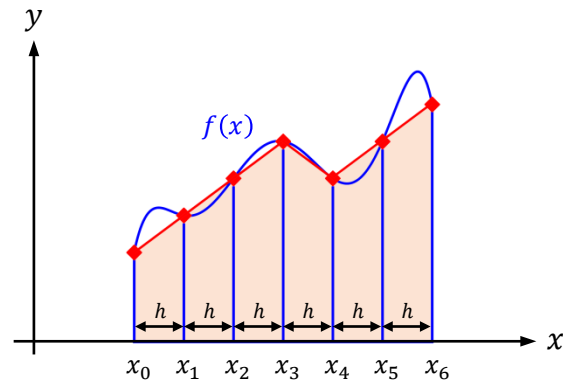
Newton-Cotes Formulas of Closed Type

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$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$R_{[a,b]}^{(1)} = R_{[x_0,x_1]}^{(1)} + R_{[x_1,x_2]}^{(1)} + \cdots + R_{[x_{N-1},x_N]}^{(1)} = -\frac{h^3}{12} \cdot f''(\xi_1) - \frac{h^3}{12} \cdot f''(\xi_2) - \cdots - \frac{h^3}{12} \cdot f''(\xi_N);$$



$$\begin{aligned} x_0 &\leq \xi_1 \leq x_1 \\ x_1 &\leq \xi_2 \leq x_2 \\ &\vdots \\ x_{N-1} &\leq \xi_N \leq x_N \end{aligned}$$

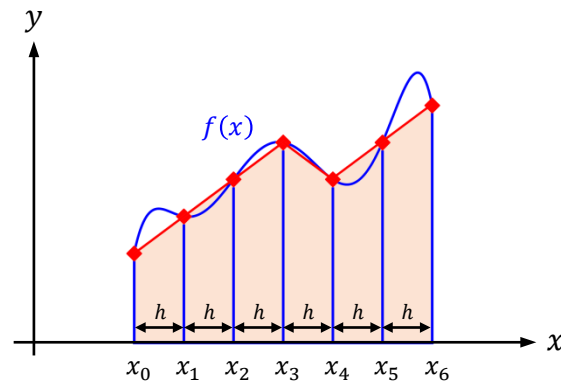
Newton-Cotes Formulas of Closed Type

The Trapezoidal Rule: The Global Truncation Error

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$$\begin{aligned} R_{[a,b]}^{(1)} &= R_{[x_0,x_1]}^{(1)} + R_{[x_1,x_2]}^{(1)} + \cdots + R_{[x_{N-1},x_N]}^{(1)} = -\frac{h^3}{12} \cdot f''(\xi_1) - \frac{h^3}{12} \cdot f''(\xi_2) - \cdots - \frac{h^3}{12} \cdot f''(\xi_N); \\ &= -\frac{N \cdot h^3}{12} \cdot f''(\xi); \quad x_0 \leq \xi \leq x_N \end{aligned}$$



$$\begin{aligned} x_0 &\leq \xi_1 \leq x_1 \\ x_1 &\leq \xi_2 \leq x_2 \\ &\vdots \\ x_{N-1} &\leq \xi_N \leq x_N \end{aligned}$$

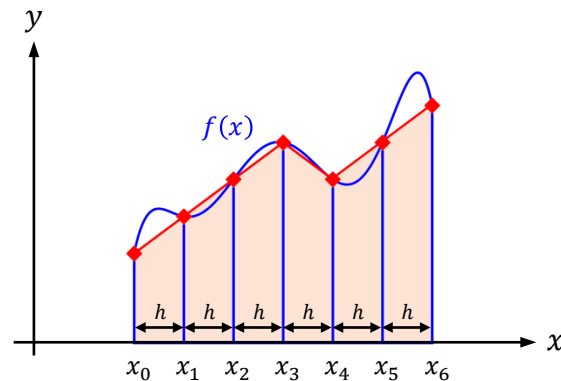
Newton-Cotes Formulas of Closed Type

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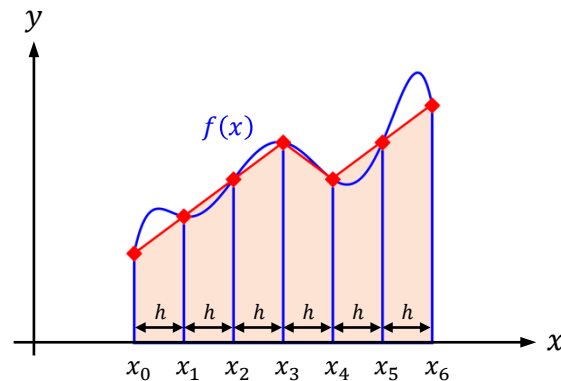
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